

Meta-analytic structural equation modeling as a framework for estimating between- and within-individual effects in multilevel dynamic models

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Abstract

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Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis, mixed-effects meta-analysis, distal outcomes

Empirical Example

To illustrate the utility of the MetaVAR framework, we applied it to data on adolescents' affective dynamics and substance use. Substance use during adolescence remains a major public health concern, contributing to morbidity, mortality, and long-term psychosocial difficulties (Patrick et al., 2023). Early onset of substance use is associated with elevated risk for dependence, cognitive impairment, and behavioral problems that persist into adulthood (Chassin et al., 2009; Squeglia et al., 2009). Youth in socioeconomically disadvantaged contexts are particularly vulnerable because they experience heightened exposure to chronic stress, limited access to adaptive coping resources, and greater social

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This research was made possible by the Prevention and Methodology Training Program (PAMT) funded by a T32 training grant (T32 DA017629 MPIs: J. Maggs & S. Lanza) from the National Institute on Drug Abuse (NIDA).

Your grants and grants that supported the miLife project.

Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer.

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reinforcement of risk-taking behaviors (Leventhal & Brooks-Gunn, 2000; Wills et al., 2004). Understanding the affective mechanisms that contribute to these disparities in risk remains a key goal for prevention science.

Affective Processes and Substance Use

Affective processes are central to theories of adolescent substance use. Two complementary motivational models, namely **negative reinforcement** (substances used to alleviate negative affect; Baker et al., 2004) and **positive reinforcement** (substances used to enhance or prolong positive affect; Cooper et al., 1995), suggest that substance use functions as an emotion-regulation strategy. However, empirical findings linking affect and substance use are often inconsistent (Hussong et al., 2011; Simons et al., 2010), perhaps because most studies focus on average levels of affect rather than dynamic processes that unfold over time. For adolescents, whose affective systems are still developing and particularly reactive to contextual stressors (Casey et al., 2008), examining the temporal organization of affect may reveal how emotion regulation contributes to early risk for substance use and behavioral problems.

Recent ecological momentary assessment research also highlights how modern contexts shape these emotion-regulation processes. For instance, in the *miLife* study, adolescents who spent more time using digital technology reported greater same-day ADHD and conduct-problem symptoms and showed increased conduct problems 18 months later (George et al., 2017). These findings suggest that heightened engagement with digital media may both reflect and exacerbate difficulties in affective and behavioral regulation, paralleling the emotion-regulation mechanisms proposed in models of substance use.

Affective Dynamics as Emotion Regulation Processes

Emerging research highlights that affective dynamics, or how emotions persist and influence one another over time, offer critical insight into emotion regulation and psychopathology (Houben et al., 2015; Kuppens, Allen, & Sheeber, 2010). Within-person **autoregressive (AR)** effects capture emotional inertia, the extent to which current emotional states carry over to the next time point. High AR in negative affect (NA) indicates slow recovery from distress, whereas high AR in positive affect (PA) reflects sustained positive mood and resilience (Hollenstein, 2015). **Cross-regressive (CR)** effects capture coupling between affective systems, or how one emotion predicts changes in another (Bringmann et al., 2016; Chow et al., 2011). For instance, a strong negative NA→PA effect indicates poor recovery from negative states, whereas a strong negative PA→NA effect reflects effective emotional buffering.

Recent ecological momentary assessment (EMA) studies further show that adolescents' daily affective responses to stressors fluctuate systematically and have meaningful behavioral correlates. For example, using the *miLife* dataset, Odgers and Russell (2017) found that on days adolescents were exposed to violence, they experienced elevated anger, depression, and conduct-problem symptoms, and that depressive symptoms often persisted into the next day. Importantly, adolescents who exhibited stronger same-day behavioral reactivity to violence were more likely to engage in substance use at follow-up, underscoring

that the temporal organization of affect is a key marker of emotion-regulation capacity and future behavioral risk.

The **stable mean attractor** complements AR and CR by representing the emotional equilibrium, or “set point,” to which affective states tend to return (Hollenstein, 2015; Kuppens, Oravecz, & Tuerlinckx, 2010). A higher attractor for NA reflects a chronically dysphoric baseline associated with stress vulnerability, whereas a higher attractor for PA reflects positive emotional homeostasis and resilience. Together, AR, CR, and attractor parameters index the persistence, interdependence, and homeostatic regulation of affect, which are dynamic markers of emotion regulation capacity relevant to adolescent health and behavior. Early applications of dynamical systems modeling to psychopathology have demonstrated the value of these approaches for capturing meaningful within-person variability. For example, Odgers et al. (2009) used a linear oscillator model to quantify the speed and regulation of weekly symptom fluctuations among high-risk psychiatric patients. Individuals showing rapid, amplifying oscillations—reflecting poor emotional damping—were more likely to engage in violent behavior, underscoring that dynamic indices of regulation capture unique aspects of risk beyond static symptom levels. Such findings illustrate how dynamic parameters like oscillation speed and stability parallel AR, CR, and attractor processes in affective systems, reinforcing their utility for studying emotion regulation in adolescence.

Affective Dynamics in Developmental Context

From a developmental and contextual perspective, affective dynamics may be particularly consequential for adolescents growing up in low-SES environments, where chronic exposure to environmental stressors (e.g., violence, financial strain, instability) taxes emotion regulation systems (Brody et al., 2013; Evans & Kim, 2012). Persistent negative affect and weak cross-affective recovery may promote substance use as a means of coping, consistent with the negative reinforcement model, whereas greater positive-affect persistence and effective PA→NA dampening may confer resilience. These parameters represent individualized dynamic signatures of emotion regulation capacity (Hamaker et al., 2018), distinguishing adolescents at greater versus lower risk for later substance use and behavioral problems. Extending this framework, findings from the *miLife* study indicate that adolescents were more likely to engage in antisocial behavior on days when they witnessed others using substances, both inside and outside the home (Russell et al., 2015). Such daily exposure to substance-use contexts may amplify emotional reactivity and dysregulation, providing a proximal pathway through which contextual stressors shape adolescents’ affective dynamics and behavioral risk.

In addition to objective socioeconomic adversity, adolescents’ perceptions of their own social standing also shape emotional and behavioral functioning. In the *miLife* study, lower subjective social status predicted higher daily symptoms of depression, anxiety, and inattention or impulsivity, as well as greater risk for substance use 18 months later, even after accounting for objective socioeconomic indicators (Russell & Odgers, 2019). These findings highlight that perceived social position within the social hierarchy may influence how adolescents experience and regulate affect in daily life, suggesting that both structural disadvantage and subjective social evaluations jointly shape affective dynamics and developmental risk.

Family Covariates Shaping Affective Dynamics

Adolescents' affective dynamics are shaped not only by family processes but also by their immediate daily contexts. For instance, exposure to substance-use environments has been linked to same-day increases in antisocial behavior among at-risk youth (Russell et al., 2015), suggesting that contextual stressors and parental monitoring jointly influence emotional and behavioral regulation. These findings highlight that adolescents' emotion regulation capacities develop within a network of overlapping influences from both their environments and their families.

Family processes form the proximal context through which adolescents learn emotional self-regulation skills. Parenting practices, particularly **parental monitoring** and **parent-child communication**, play a critical role in providing structure, predictability, and opportunities for emotion socialization (Eisenberg et al., 2010; Morris et al., 2007). High parental monitoring promotes emotional stability by reducing adolescents' exposure to risky contexts and helping them manage distress (Dishion & McMahon, 1998; Stattin & Kerr, 2000). Conversely, low monitoring and inconsistent supervision may lead to greater emotional reactivity and difficulty recovering from negative affect. Similarly, open and supportive communication fosters positive emotional expression and flexibility, whereas poor communication may result in suppressed PA and prolonged NA.

These effects may be especially pronounced in disadvantaged contexts, where stress undermines parenting consistency (Brody et al., 2013; Evans & Kim, 2012; Leventhal & Brooks-Gunn, 2000). Adolescents experiencing high parental monitoring and warmth are expected to show adaptive affective dynamics, such as faster recovery from NA (lower AR[NA]), stronger PA→NA dampening, and more positive affective attractors, whereas those from low-monitoring, low-communication homes may exhibit maladaptive dynamics such as higher inertia and greater cross-affective spillover. Modeling parental monitoring and communication as covariates predicting person-specific affective parameters allows us to test whether early family environments shape how emotions fluctuate, persist, and recover in daily life.

Affective Dynamics as Mediating Mechanisms

Beyond direct associations, affective dynamics may mediate pathways from family context to adolescent behavioral outcomes. Supportive parenting may cultivate adaptive affective patterns, such as stable positive mood, faster recovery from distress, and more positive attractors, which in turn reduce risk for substance use and externalizing problems. Conversely, inconsistent monitoring and poor communication may foster maladaptive dynamics, including high NA inertia, negative PA→NA coupling, and dysphoric attractors, that increase reliance on substances for affect regulation. Evidence from ecological momentary assessment supports these propositions: daily affective reactivity to environmental stressors predicts both immediate and long-term outcomes. In the *miLife* study, adolescents who reacted more strongly to violence exposure in daily life were at greater risk for later substance use (Odgers & Russell, 2017). This evidence suggests that moment-to-moment emotion regulation processes may serve as a key mechanism through which family and contextual factors “get under the skin” to shape adolescents' emotional and behavioral trajectories.

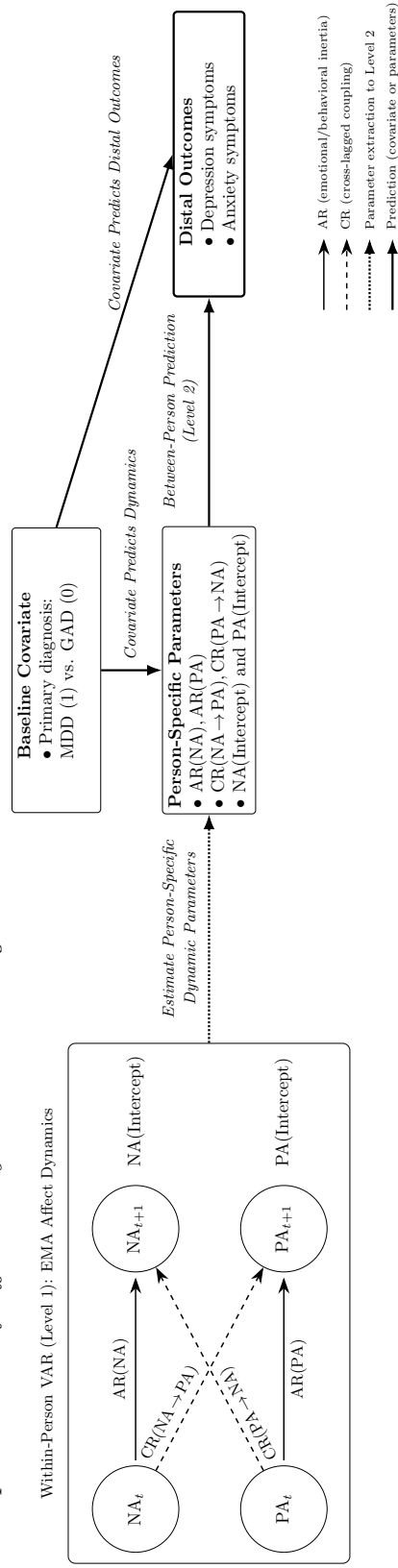
The MetaVAR Framework as an Integrative Approach

The **meta-analytic vector autoregressive (MetaVAR)** framework provides a unifying analytic structure for testing these developmental hypotheses. MetaVAR simultaneously estimates person-specific AR, CR, and attractor parameters from intensive longitudinal data (for example, daily EMA) and links them to contextual covariates and distal outcomes. This multilevel modeling framework accommodates both within-person temporal processes and between-person predictors, allowing researchers to (a) quantify individual differences in affective dynamics, (b) test how parenting and contextual factors influence those dynamics, and (c) evaluate how these emotion-regulation patterns predict future substance use and behavioral problems.

In the present study, we apply the MetaVAR framework to data from the *miLife Study* (Odgers & Russell, 2017; Russell et al., 2015), a longitudinal ecological momentary assessment (EMA) investigation that followed 151 adolescents (ages 11-15) from socioeconomically disadvantaged neighborhoods. Participants completed brief mobile-phone assessments three times per day for 30 consecutive days, yielding more than 12,000 assessments across 4,000 person-days. This intensive design captured adolescents' daily affect, contextual stressors, and behavioral experiences in real time, minimizing recall bias and enabling fine-grained modeling of within-person affective dynamics. The study was designed to examine how contextual risk factors such as exposure to violence, witnessing substance use, and daily stress shape adolescents' emotional and behavioral functioning. Prior analyses from the *miLife* dataset have shown that daily fluctuations in affect and exposure predict same-day and next-day changes in behavior and mental health symptoms (Odgers & Russell, 2017; Russell et al., 2015), making it ideally suited for testing hypotheses about affective dynamics and their developmental consequences. This approach enables a comprehensive test of an integrative developmental model in which early family environments shape adolescents' day-to-day affective dynamics—how emotions persist, interact, and recover in real time—which, in turn, forecast later substance use and behavioral outcomes. The *miLife* data provide an especially strong foundation for this analysis because they capture both proximal (daily) emotional processes and distal (18-month follow-up) behavioral outcomes within the same cohort (Odgers & Russell, 2017; Russell et al., 2015). Thus, the MetaVAR framework can bridge within-person emotion-regulation dynamics and between-person developmental trajectories among socioeconomically disadvantaged youth (Figure 1). The purpose of this empirical example is to demonstrate how the MetaVAR approach estimates individual differences in affective dynamics and links them to contextual covariates and behavioral outcomes, illustrating the framework's potential for developmental and clinical applications.

Research Questions and Hypotheses

Building on prior theory and empirical findings on affective dynamics and adolescent development, we pose the following research questions (RQs) and hypotheses (Hs) to guide the current investigation.

Figure 1*Conceptual Model of Affective Dynamics Predicting Adolescent Outcomes*

RQ1: Characterizing and Decomposing Affective Dynamics

- **RQ1a.** What are the typical within-person dynamics of positive affect (PA) and negative affect (NA) in adolescents, as captured by autoregressive (AR), cross-regressive (CR), and attractor parameters?
- **RQ1b.** To what extent do these within-person affective dynamics vary across individuals (i.e., between-person variability in AR, CR, and attractor parameters)?
- **RQ1c.** How strongly are the within-person dynamic parameters correlated at the between-person level (e.g., do adolescents with higher NA inertia also show weaker PA→NA recovery)?

H1. On average, adolescents will exhibit moderate positive autoregression (emotional inertia) for both PA and NA, negative cross-regression (PA→NA), and emotional attractors centered near affective balance. However, substantial between-person variability is expected in all dynamic parameters, reflecting individual differences in emotion regulation capacity. Specifically, adolescents who recover more slowly from distress (higher AR[NA]) are expected to exhibit weaker PA→NA coupling and higher NA attractors, indicating more dysphoric equilibrium states.

RQ2: Family Processes and Affective Dynamics

- **RQ2a.** Does parental monitoring predict individual differences in the persistence and coupling of affective states?
- **RQ2b.** Does parent–child communication predict individual differences in affective attractors and cross-affective recovery?

H2.1. Higher parental monitoring will be associated with lower AR(NA) (faster recovery from distress), stronger negative PA→NA coupling (effective emotional dampening), and higher PA attractors (more positive equilibrium).

H2.2. Poor monitoring and communication will predict higher NA inertia and higher NA attractors, reflecting chronic dysphoria and stress vulnerability.

H2.3. Supportive communication will predict higher AR(PA) (sustained positive emotion) and more adaptive cross-affective coupling (greater emotional flexibility).

RQ3: Affective Dynamics as Mediators of Family Effects on Substance Use

- **RQ3a.** Do adolescents from high-monitoring and high-communication families exhibit affective dynamics that buffer against later substance use?
- **RQ3b.** Do adolescents from low-monitoring and low-communication families exhibit maladaptive affective dynamics that increase risk for substance use?

H3.1. Adaptive affective dynamics (low NA inertia, strong PA→NA dampening, high PA attractor) will mediate the protective effects of parental monitoring and communication on later substance use.

H3.2. Maladaptive affective dynamics (high NA inertia, weak PA→NA coupling, high NA attractor) will mediate the risk-enhancing effects of poor family processes on later substance use.

References

- Baker, T. B., Piper, M. E., McCarthy, D. E., Majeskie, M. R., & Fiore, M. C. (2004). Addiction motivation reformulated: An affective processing model of negative reinforcement. *Psychological Review*, 111(1), 33–51. <https://doi.org/10.1037/0033-295x.111.1.33>
- Bringmann, L. F., Pe, M. L., Vissers, N., Ceulemans, E., Borsboom, D., Vanpaemel, W., Tuerlinckx, F., & Kuppens, P. (2016). Assessing temporal emotion dynamics using networks. *Assessment*, 23(4), 425–435. <https://doi.org/10.1177/1073191116645909>
- Brody, G. H., Yu, T., Chen, E., Miller, G. E., Kogan, S. M., & Beach, S. R. H. (2013). Is resilience only skin deep?: Rural African Americans' socioeconomic status-related risk and competence in preadolescence and psychological adjustment and allostatic load at age 19. *Psychological Science*, 24(7), 1285–1293. <https://doi.org/10.1177/0956797612471954>
- Casey, B., Jones, R. M., & Hare, T. A. (2008). The adolescent brain. *Annals of the New York Academy of Sciences*, 1124(1), 111–126. <https://doi.org/10.1196/annals.1440.010>
- Chassin, L., Hussong, A., & Beltran, I. (2009). Adolescent substance use. In R. M. Lerner & L. Steinberg (Eds.), *Handbook of adolescent psychology: Individual bases of adolescent development* (3rd ed., pp. 723–763). Wiley. <https://doi.org/10.1002/9780470479193.adlpsy001022>
- Chow, S.-M., Zu, J., Shifren, K., & Zhang, G. (2011). Dynamic factor analysis models with time-varying parameters. *Multivariate Behavioral Research*, 46(2), 303–339. <https://doi.org/10.1080/00273171.2011.563697>
- Cooper, M. L., Frone, M. R., Russell, M., & Mudar, P. (1995). Drinking to regulate positive and negative emotions: A motivational model of alcohol use. *Journal of Personality and Social Psychology*, 69(5), 990–1005. <https://doi.org/10.1037/0022-3514.69.5.990>
- Dishion, T. J., & McMahon, R. J. (1998). Parental monitoring and the prevention of child and adolescent problem behavior: A conceptual and empirical formulation. *Clinical Child and Family Psychology Review*, 1(1), 61–75. <https://doi.org/10.1023/a:1021800432380>
- Eisenberg, N., Spinrad, T. L., & Eggum, N. D. (2010). Emotion-related self-regulation and its relation to children's maladjustment. *Annual Review of Clinical Psychology*, 6(1), 495–525. <https://doi.org/10.1146/annurev.clinpsy.121208.131208>
- Evans, G. W., & Kim, P. (2012). Childhood poverty, chronic stress, self-regulation, and coping. *Child Development Perspectives*, 7(1), 43–48. <https://doi.org/10.1111/cdep.12013>
- George, M. J., Russell, M. A., Piontak, J. R., & Odgers, C. L. (2017). Concurrent and subsequent associations between daily digital technology use and high-risk adolescents' mental health symptoms. *Child Development*, 89(1), 78–88. <https://doi.org/10.1111/cdev.12819>
- Hamaker, E. L., Asparouhov, T., Brose, A., Schmiedek, F., & Muthén, B. (2018). At the frontiers of modeling intensive longitudinal data: Dynamic structural equation models for the affective measurements from the COGITO study. *Multivariate Behavioral Research*, 53(6), 820–841. <https://doi.org/10.1080/00273171.2018.1446819>

- Hollenstein, T. (2015). This time, it's real: Affective flexibility, time scales, feedback loops, and the regulation of emotion. *Emotion Review*, 7(4), 308–315. <https://doi.org/10.1177/1754073915590621>
- Houben, M., Van Den Noortgate, W., & Kuppens, P. (2015). The relation between short-term emotion dynamics and psychological well-being: A meta-analysis. *Psychological Bulletin*, 141(4), 901–930. <https://doi.org/10.1037/a0038822>
- Hussong, A. M., Jones, D. J., Stein, G. L., Baucom, D. H., & Boeding, S. (2011). An internalizing pathway to alcohol use and disorder. *Psychology of Addictive Behaviors*, 25(3), 390–404. <https://doi.org/10.1037/a0024519>
- Kuppens, P., Allen, N. B., & Sheeber, L. B. (2010). Emotional inertia and psychological maladjustment. *Psychological Science*, 21(7), 984–991. <https://doi.org/10.1177/0956797610372634>
- Kuppens, P., Oravecz, Z., & Tuerlinckx, F. (2010). Feelings change: Accounting for individual differences in the temporal dynamics of affect. *Journal of Personality and Social Psychology*, 99(6), 1042–1060. <https://doi.org/10.1037/a0020962>
- Leventhal, T., & Brooks-Gunn, J. (2000). The neighborhoods they live in: The effects of neighborhood residence on child and adolescent outcomes. *Psychological Bulletin*, 126(2), 309–337. <https://doi.org/10.1037/0033-2909.126.2.309>
- Morris, A. S., Silk, J. S., Steinberg, L., Myers, S. S., & Robinson, L. R. (2007). The role of the family context in the development of emotion regulation. *Social Development*, 16(2), 361–388. <https://doi.org/10.1111/j.1467-9507.2007.00389.x>
- Odgers, C. L., Mulvey, E. P., Skeem, J. L., Gardner, W., Lidz, C. W., & Schubert, C. (2009). Capturing the ebb and flow of psychiatric symptoms with dynamical systems models. *American Journal of Psychiatry*, 166(5), 575–582. <https://doi.org/10.1176/appi.ajp.2008.08091398>
- Odgers, C. L., & Russell, M. A. (2017). Violence exposure is associated with adolescents' same- and next-day mental health symptoms. *Journal of Child Psychology and Psychiatry*, 58(12), 1310–1318. <https://doi.org/10.1111/jcpp.12763>
- Patrick, M., Miech, R., Johnston, L., & O'Malley, P. (2023). *Monitoring the Future Panel Study annual report: National data on substance use among adults ages 19 to 60, 1976-2022*. Ann Arbor, MI, Institute for Social Research, The University of Michigan. <https://doi.org/10.7826/isr-um.06.585140.002.07.0002.2023>
- Russell, M. A., & Odgers, C. L. (2019). Adolescents' subjective social status predicts day-to-day mental health and future substance use. *Journal of Research on Adolescence*, 30(S2), 532–544. <https://doi.org/10.1111/jora.12496>
- Russell, M. A., Wang, L., & Odgers, C. L. (2015). Witnessing substance use increases same-day antisocial behavior among at-risk adolescents: Gene-environment interaction in a 30-day ecological momentary assessment study. *Development and Psychopathology*, 28(4pt2), 1441–1456. <https://doi.org/10.1017/s0954579415001182>
- Simons, J. S., Dvorak, R. D., Batien, B. D., & Wray, T. B. (2010). Event-level associations between affect, alcohol intoxication, and acute dependence symptoms: Effects of urgency, self-control, and drinking experience. *Addictive Behaviors*, 35(12), 1045–1053. <https://doi.org/10.1016/j.addbeh.2010.07.001>

- Squeglia, L. M., Jacobus, J., & Tapert, S. F. (2009). The influence of substance use on adolescent brain development. *Clinical EEG and Neuroscience*, 40(1), 31–38. <https://doi.org/10.1177/155005940904000110>
- Stattin, H., & Kerr, M. (2000). Parental monitoring: A reinterpretation. *Child Development*, 71(4), 1072–1085. <https://doi.org/10.1111/1467-8624.00210>
- Wills, T. A., Resko, J. A., Ainette, M. G., & Mendoza, D. (2004). Role of parent support and peer support in adolescent substance use: A test of mediated effects. *Psychology of Addictive Behaviors*, 18(2), 122–134. <https://doi.org/10.1037/0893-164x.18.2.122>